Chapter 4 — Cross-Chain MEV between Solana and EVM: An End-to-End Empirical Framework on Wormhole Portal

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1 WHY WORMHOLE PORTAL, AND HOW THE DATASET WAS BUILT

1.1 Why Wormhole Portal?

Five general-purpose bridges currently dominate the Solana↔Ethereum corridor: Relay, Mayan Finance, Chainlink CCIP, LayerZero V2, and Wormhole Portal. Wormhole was chosen as the exclusive subject of this study for five interlocking reasons.

(1) **Longest continuous history and deepest traffic.** As the earliest permissionless generalized messaging layer between the two ecosystems, Portal has the longest contiguous event record (the `/operations` API returns every cross-chain message back-indexed), and carries the deepest daily SOL↔ETH transfer flow among all five bridges, yielding the statistical power needed for a 365-day study.

(2) **Direct on-chain verifiability through VAAs.** Every Wormhole message is cryptographically signed by the Guardian set and written as a *Verifiable Action Approval* (VAA). This makes it possible to tie one Solana signature unambiguously to one Ethereum transaction hash without relying on off-chain bookkeeping, which is essential for cross-chain PnL reconstruction.

(3) **Stable pair schema.** Portal's Token Bridge contracts on both chains use a fixed emitter/recipient schema, so the payer and receiver on each side can be decoded uniformly for every record. This permits us to build the (sol_addr, eth_addr) *actor* key that underlies every concentration, round-trip, and actor-feature analysis.

(4) **Clean separation between transport and execution.** The bridge executes only the transfer primitive; any subsequent DEX execution is an independent on-chain event with its own logs. This means one can isolate the three sub-processes — bridge leg, entry swap leg, exit swap leg — without being entangled with the bridge's own accounting.

(5) **Token coverage that matches the observed arbitrage universe.** The empirically observed arbitrage universe is dominated by long-tail memecoins, and Wormhole (through both the WTT and NTT standards) supports a strict superset of what any of the other four bridges carry in this segment. Restricting to Wormhole therefore does *not* induce meaningful selection bias on the phenomena studied.

1.2 Data-Collection Pipeline

We collect data in five sequential stages (Figure 1). The raw volume of the pipeline is substantial: $\sim$ 30,729 candidate arbitrage records, $\sim$ 292 MB of matched-context JSON, and ~ 804 MB of per-token Ethereum-side 1-minute price series. Since parsing these in memory repeatedly caused OS-level failures on the test workstation, the pipeline was re-engineered with module-level singleton caches, per-token price sharding (165 LRU-managed files), and five disk checkpoints for the predictive-model sub-pipeline. The full pipeline now runs end-to-end in $<$ 2 GB of resident memory.

Walk-through of Figure 1. The diagram reads left-to-right as four labeled stages. **Stage 1 (Raw sources)** is the left-most column of three coloured boxes: the Wormhole `/operations` API (yellow), Solana/Ethereum Portal contract logs (blue), and Birdeye/DeFiLlama 1-minute price grids (green). **Stage 2 (Pair matching)** is the single middle box that joins the two per-chain signature streams into one cross-chain record via the VAA chain-ID + sequence number, producing the 4-tuple (`direction`, `sol_sig`, `eth_sig`, `eth_hash`). **Stage 3 (Enrichment)** is a three-box column that extracts arbitrage actors (`sol_addr, eth_addr`), attaches swap legs via Helius `getTx` + Etherscan, and pulls per-token prices from a 165-file sharded LRU cache. **Stage 4 (Output artefacts)** is the right-most two boxes: the reconstructed PnL ledger (`entry_cost`, `exit_value`, `gross/net_pnl`, ROI) on top and the 17-dimensional feature frame used by the downstream risk-model validation on the bottom. Arrows in between carry explicit labels (*price lookup*, *actors*, *swaps*, *prices*) so the reader can trace which artefact feeds which stage. Note: the working-sample funnel widths (30,729 → 28,574 → 28,554) are reported in the text below, not printed on the diagram itself.

Stage 1 — Raw sources. For the window 2025-04-15 — 2026-04-14 (364 days), we query three raw data sources in parallel: (i) the Wormhole `/operations` API to enumerate every SOL↔ETH message; (ii) Helius RPC `getSignaturesForAddress` for Solana Portal contract logs; (iii) Etherscan / direct RPC for Ethereum Portal contract logs. In parallel we pre-fetch a 1-minute price grid from Birdeye for Solana-side tokens and from DeFiLlama for Ethereum-side tokens.

Stage 2 — Pair matching. Each Wormhole message exposes a VAA chain-ID and a sequence number, which uniquely identifies the complementary leg on the other chain. We match every source-leg signature `sol_sig` to its destination-leg hash `eth_hash` (or vice versa for the opposite direction). Matching is pure-cryptographic: we do *not* rely on time-window heuristics. After matching, each record has the 4-tuple (`direction, sol_sig, eth_sig, eth_hash`) and the Wormhole-emitted amounts on each side.

Stage 3 — Actor extraction and context enrichment. From every matched record we decode the signer/payer fields on both sides, giving two *actor* attributes: `arb_sol` (Solana base58 pubkey) and `arb_eth` (Ethereum hex address). We then union these across all records, obtaining a global actor set of 321 unique (sol_addr, eth_addr) pairs. For every actor we additionally pull their full 1-year transaction history on both chains, so that each candidate event can be placed inside its local DEX-activity context: the swaps that occurred on the actor's wallet within a configurable window (typically ±60 s) of the bridge leg are extracted as `entry_swaps_greedy` and `exit_swaps_greedy`. This is the step that distinguishes a *bridge transfer* from a *cross-chain arbitrage*: the latter requires *both* legs to have a nearby DEX swap.

Stage 4 — PnL reconstruction. For each candidate we compute:

- $C^{\text{in}}_{\text{exec}}$: the realized USD cost of the entry leg, by multiplying each input-token amount by its price at `sol_ts` (or `eth_ts`) from the per-token 1-minute price grid;
- V^{out}: the realized USD value of the exit leg, using the exit-side token price at exit timestamp;
- $\Pi_{\text{gross}} = V^{\text{out}} - C^{\text{in}}_{\text{exec}}$;
- $\Pi_{\text{net}} = \Pi_{\text{gross}} - f_{\text{sol}} - f_{\text{eth}}$, subtracting the realized per-chain gas fees.

Dust-entry artefact filter. Before any downstream analysis, we drop candidates whose $C^{\text{in}}_{\text{exec}}$ falls below \$1, below the minimum plausible on-chain trade cost (an Ethereum L1 swap alone costs \$1–\$5 in gas). These rows are pipeline artefacts: token-amount rounding in bridge wrapping collapses the denominator of ROI = $\Pi_{\text{net}}/C^{\text{in}}_{\text{exec}}$, producing values up to 10^{11}% (the five most extreme are all ETH→SOL TRUMP transfers with $C^{\text{in}}_{\text{exec}} < 10^{-5}$ USD and PnL in the hundreds of dollars). The filter removes 11 rows (0.04%); we intentionally do *not* filter on *fee > entry*, which is a legitimate loss rather than a measurement error. After this step the candidate set contains 30,729 events.

A reliability flag `pnl_reliability` ∈ {reliable, reliable_after_dedup_failed, unreliable_asymmetric, unreliable_partial_holdings} is assigned based on leg-coverage sanity checks. This study uses the `reliable` tier (N = 28,574; 93.0% of the candidate sample), from which we further remove the 20 most extreme PnL events (top-10 worst losses and top-10 biggest gains, collectively < 0.07% of the reliable sample) whose PnL values are pricing-pipeline artefacts on essentially-empty pools (e.g. `dead_pool_exploit` / `thin_pool_arb` entries with wrapped-asset SOL-side liquidity ≤ \$25 k against the ≥ \$10 k entry sizes they represent). The working sample is therefore $N = 28{,}554$.

Stage 5 — Feature frame. The reliable subset is joined with liquidity attributes (`sol_liq_usd, eth_liq_usd`), 24-h volume proxies, market-cap estimates, and the signed entry price gap

$$g_{\text{signed}} = \text{sign}_{\text{dir}} \cdot \left(P_{\text{dst}}/P_{\text{src}} - 1\right) \times 100\%, \tag{1}$$

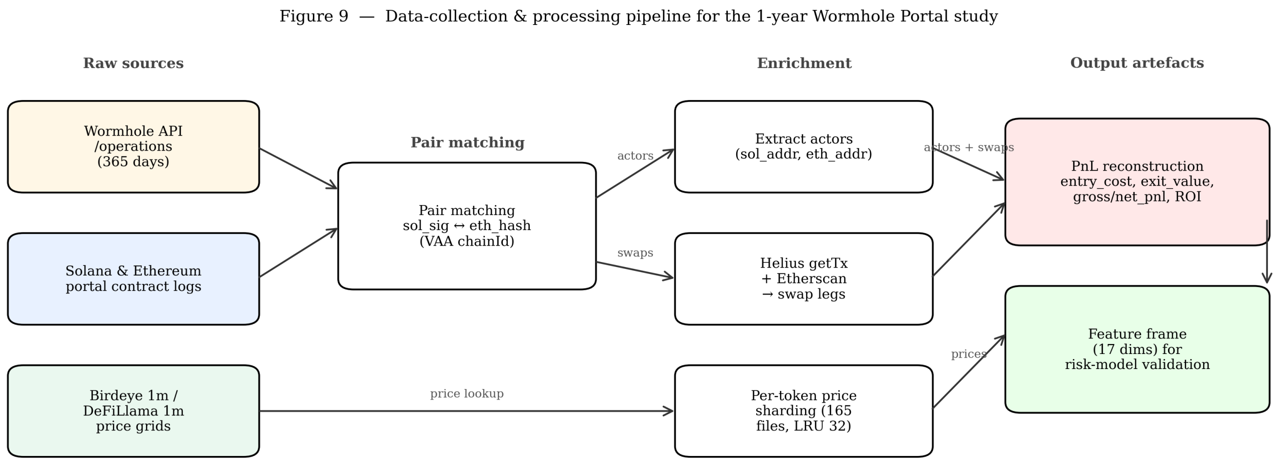

Figure 1: Data-collection and processing pipeline for the Wormhole Portal 1-year study.

computed from the *bridged-token* prices on both sides at entry time. This yields a 17-dimensional feature frame consumed by all downstream analyses.

2 CROSS-CHAIN ARBITRAGE: DEFINITION AND EMPIRICAL PROFILE

2.1 Definition

Throughout this chapter, a *cross-chain arbitrage event* is any observation e with the following structure.

> **Definition 1 (cross-chain arbitrage).** An observation e is a *cross-chain arbitrage event* if it consists of three causally linked on-chain legs
>
> $$L_{\text{entry}} \;\rightarrow\; L_{\text{bridge}} \;\rightarrow\; L_{\text{exit}}$$
>
> such that (i) L_{entry} and L_{exit} each contain at least one DEX swap on distinct chains; (ii) the transferred asset and amount of L_{bridge} are consistent with the output of L_{entry} and the input of L_{exit}; (iii) the three legs share the same (sol_addr, eth_addr) actor.

We do *not* impose that $\Pi_{\text{net}} > 0$: we are interested in the *intent* and *structure* of arbitrage, not its ex-post success. Nor do we impose that g_{signed} exceed a threshold: any observed cross-chain price dislocation, whatever its origin, is an arbitrage signal.[1] This is consistent with the earlier methodological note that "any cross-chain price divergence *is* arbitrage; the cause of the divergence and the speed of the execution are not admissibility criteria."

[1]A separate 10% gap filter is applied upstream to bound the data volume at ingestion; all subsequent analyses in this study use the full reliable subset regardless of gap magnitude.

2.2 Empirical Profile

Figures 2–7 summarize the one-year empirical profile of reliable Wormhole arbitrage.

Walk-through of Figure 2. Panel (a) is a horizontal bar chart of the `subtype` breakdown of all candidates; the title reports $N = 30{,}729$ total and $n_{\text{rel}} = 28{,}574$ reliable (93%), summarising the upstream pipeline funnel before the subsequent 20-outlier exclusion yields the working $N = 28{,}554$. Panel (b) renders $\log_{10}$ entry size by direction as side-by-side violins; both directions overlap tightly around the median ($268 SOL→ETH, $200 ETH→SOL), but SOL→ETH carries a visibly longer right tail driven by the few wrapped-asset outliers that our exclusion rule subsequently removes. Panel (c) lists the top-10 most-traded tokens as horizontal bars with absolute counts and percent labels; the token-level Herfindahl index reported in the panel title is only 665, with the bottom-most (highlighted) bar marking the single most-active token. Panel (d) is a day-of-week × hour-of-UTC heatmap: traffic peaks in Americas afternoon hours (16–22 UTC), consistent with Ethereum-side DEX rhythm, with a secondary plateau during Asia prime time. Panel (e) is the overall bridge-size histogram on log-x with the sample median marked; the distribution is log-normal with a narrow peak around $200–$300. Panel (f) is a stacked bar of (liquidity_category × direction): `thin_pool_arb` dominates (~ 16 k events), followed by `dead_pool_exploit` and `healthy_market_arb` at ~ 6 k each.

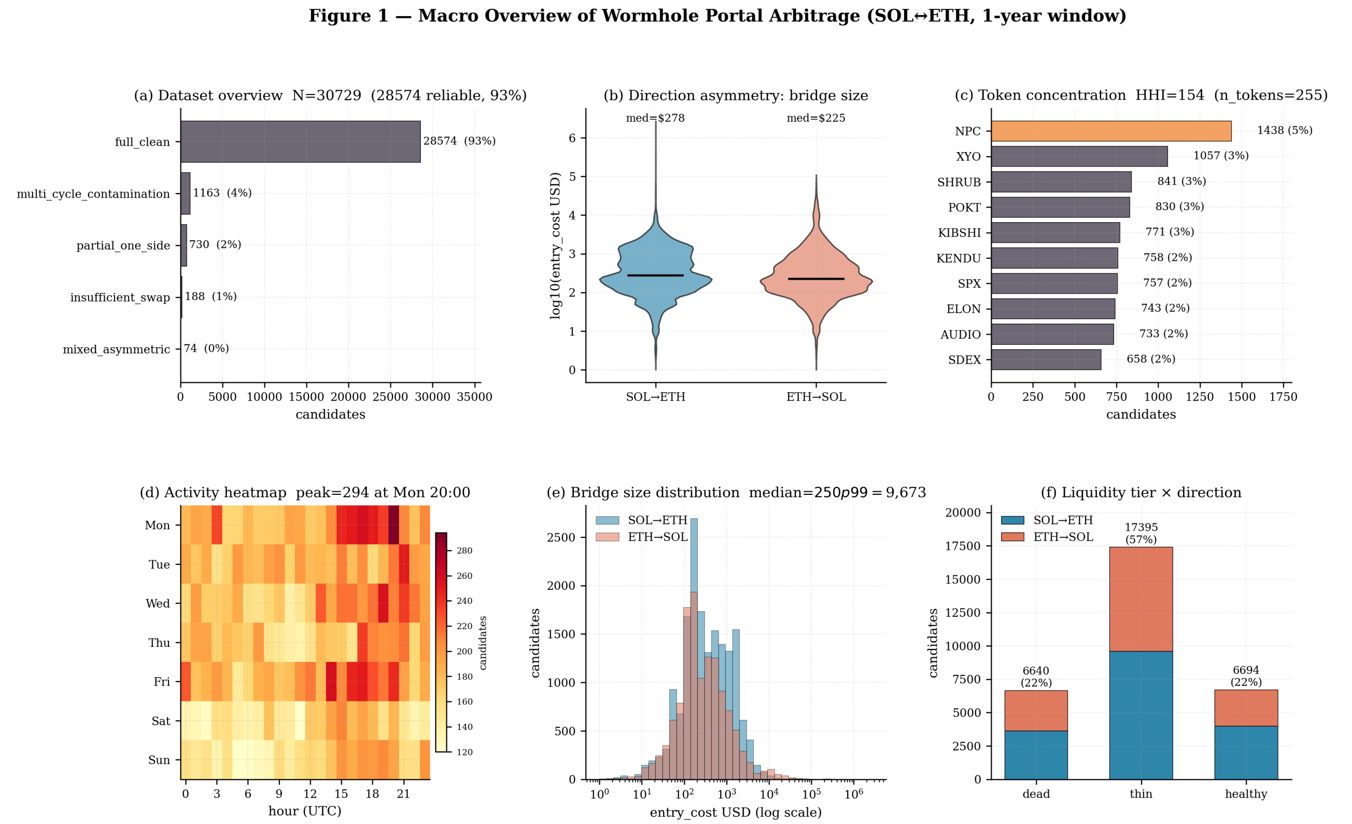

Figure 2: Macro structure: dataset funnel (a), log-size distribution by direction (b), top-15 token Herfindahl bars (c), hour-of-UTC × day-of-week activity heatmap (d), overall bridge-size histogram (e), and liquidity-tier × direction stacked bars (f).

Walk-through of Figure 3. Panel (a) plots the CDF of bridge latency $|\Delta t|$ on a log x-axis, overlaying one curve per direction: SOL→ETH converges to the 34 s plateau almost vertically, while ETH→SOL only begins plateauing above 10^3 s — the direct visual evidence for the $\approx 30\times$ latency asymmetry driven by Ethereum finality. Panel (b) is a 2×2 atomicity matrix of (entry same-block) $\times$ (exit same-block); only 28 events populate the fully-atomic cell (0.10%), with the vast mass lying in the "entry-atomic, exit-non-atomic" quadrant. Panel (c) is a histogram of $|t_{\text{swap}} - t_{\text{bridge}}|$ (log-x, clipped at 24 h) separately for entry and exit swaps: both have a sharp mode between 10 and 10^2 s, but the exit-swap distribution has a fatter right tail. Panel (d) bins observations by bridge-latency tier and draws ROI as per-tier box plots (clipped ±50%): counter-intuitively, the slowest tier carries the *highest* median ROI ($\sim$ 2–3%) while the fastest tier drops to $\sim$ 0.5%, a visual precursor of the plateau-plus-exponential $\phi(\Delta)$ fit in Section 3.

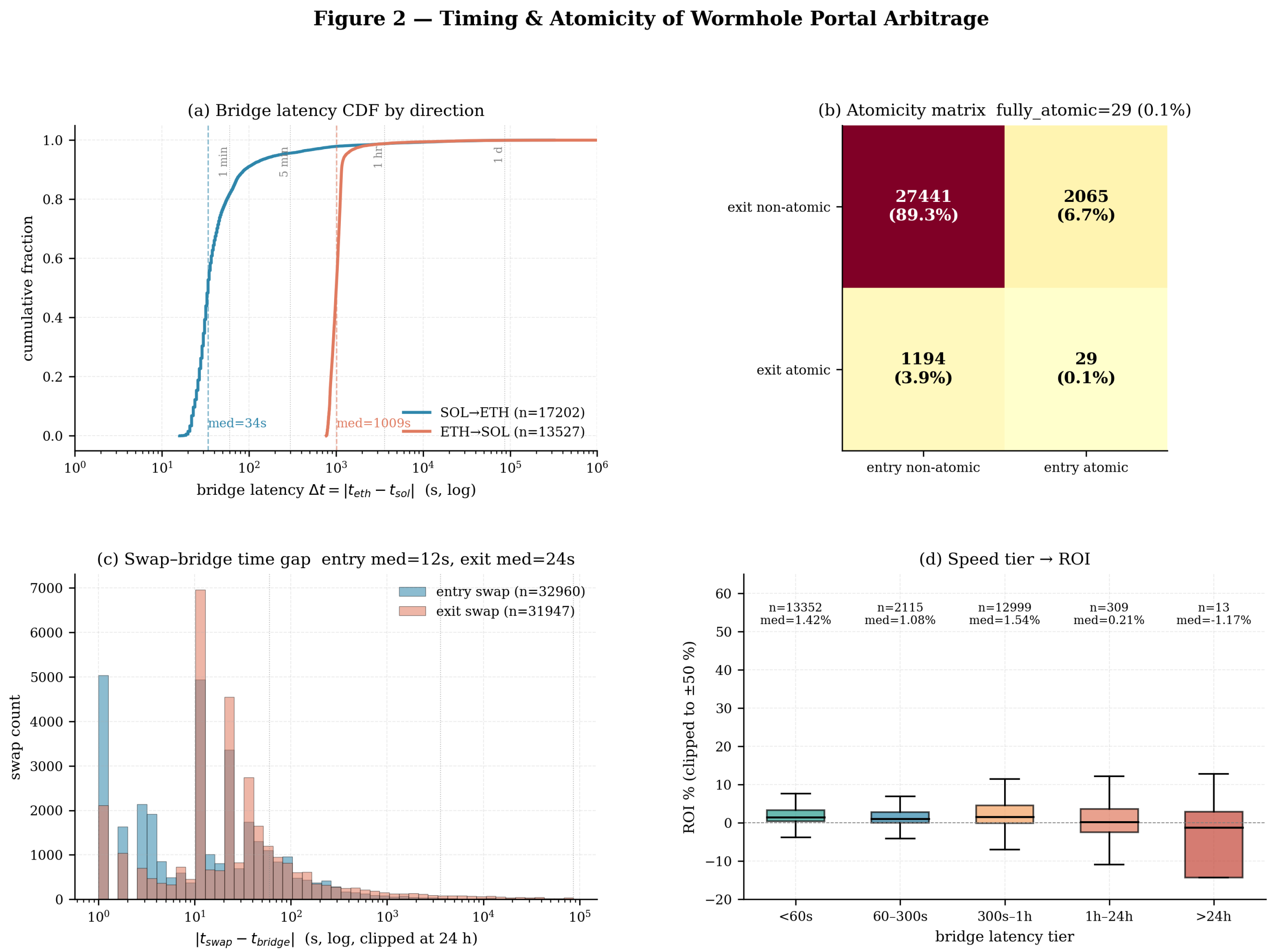

Figure 3: Time structure: bridge-latency CDF by direction (a), atomicity matrix (b), bridge-to-swap time gap (c), ROI by bridge-latency tier (d).

Walk-through of Figure 4. Panel (a) is the ROI histogram clipped to ±20% with a zoomed CDF inset over ±10%; the mass is strongly right-skewed, with a sharp lobe above 10% populated almost entirely by dead- and thin-pool events. Panel (b) overlays the two directional ROI CDFs: ETH⟶SOL has the slightly higher median (1.61%) but SOL⟶ETH carries a heavier upper tail. Panel (c) bins entry_cost into decades ($10–$100, $100–$1 k, $1 k–$10 k, $\geq$ $10 k) and draws per-tier count bars: most activity ($\sim$17 k events) lives in the $100–$1 k band, with only $\sim$ 137 events above $10 k. Panel (d) plots the fee-burden ratio $f_{\text{total}}/|\Pi_{\text{gross}}|$ on log-x; the median is well below 0.1, showing that gas is not the dominant cost on this corridor. Panel (e) is a scatter of Π_{gross} vs. Π_{net} with points colored by log(size); the points fall on or just below the 45° line, visually confirming that fees shift realized PnL by only a small constant. Panel (f) is the direct capital⟶net-PnL regression after 20-outlier exclusion: the cloud is essentially flat ($r^2 = 0.024$), overturning the earlier $r^2 = 0.93$ claim that we now know was an artefact of the removed outliers.

Figure 3 — Economics of Wormhole Portal Arbitrage (ROI, Size, Fees, PnL)

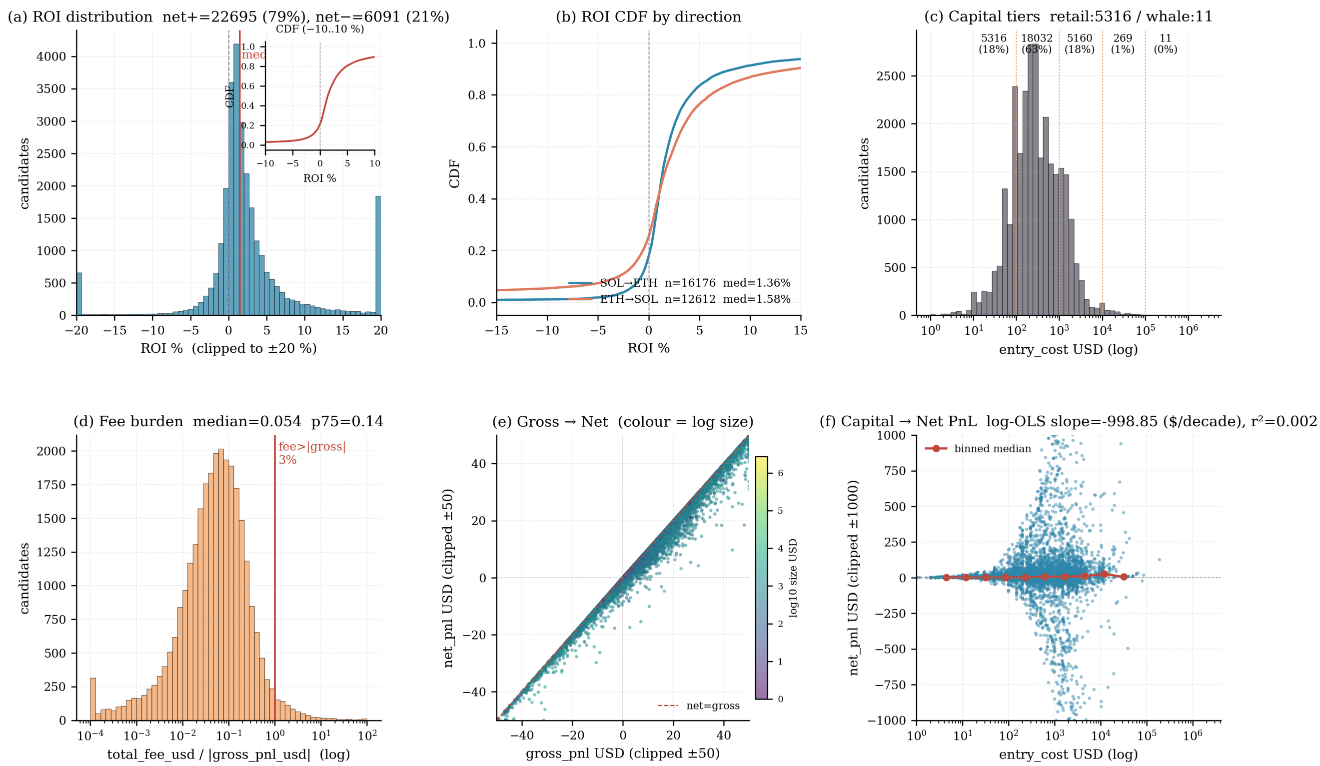

Figure 4: Economics: ROI histogram with zoomed CDF inset (a), ROI CDF by direction (b), capital-tier distribution (c), fee-burden ratio (d), gross → net PnL scatter (e), and the capital → net PnL regression (f).

Walk-through of Figure 5. Panel (a) is a grouped boxplot of ROI by (liquidity_category × direction): `dead_pool_exploit` has the *highest* median ROI ($\approx 2.3\%$) but only at sub-\$200 size; `thin_pool_arb` sits at $\approx 1.8\%$ with broader capacity; and `healthy_market_arb` caps at $\approx 0.5\%$ despite allowing \$1 k+ order sizes — the characteristic "small-trade-high-edge, large-trade-small-edge" tension. Panel (b) is a two-column plot of the bridged-token signed gap at entry (left) vs. at exit (right), connected by per-event segments; the visual message is a sharp convergence toward gap ≈ 0 at exit. Panel (c) is a scatter of entry-side g_{signed} against realized ROI, with decile-median overlays: the monotone upward trend (Spearman $\rho = 0.365$) from -0.10% at D1 to $+4.14\%$ at D10 directly validates g_{signed} as a leading signal in the risk model.

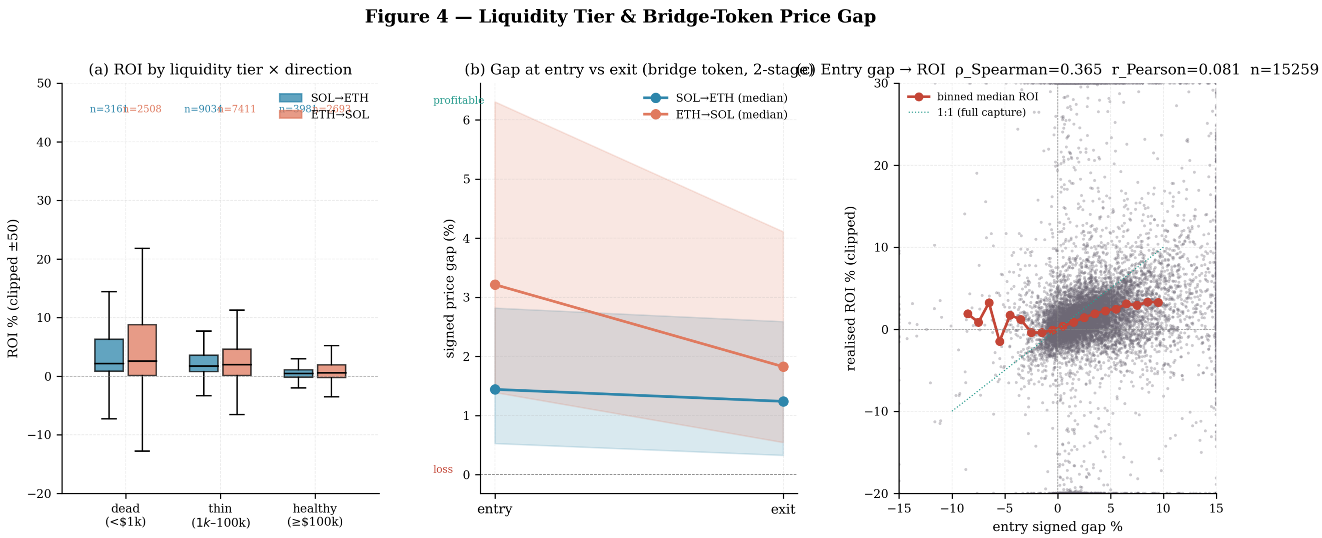

Figure 5: Liquidity and gap event study: ROI by liquidity tier × direction (a), signed-gap entry-vs.-exit two-stage visualization (b), entry gap → realized ROI with decile medians (c).

Walk-through of Figure 6. Panel (a) is the Lorenz curve of per-actor trade counts (sorted low→high): the sharp elbow in the top $\sim 5\%$ produces Gini $= 0.932$ and HHI $= 665$, quantifying an extremely skewed searcher population where the top-5 actors alone carry $\sim 50\%$ of all trades. Panel (b) is a grouped box-plot of ROI by `pnl_reliability` tier: *reliable* sits at median $\approx +1.4\%$ while *unreliable_asymmetric* collapses to median $\approx -35\%$ — the reliability flag is empirically load-bearing. Panel (c) is a scatter of round-trip events with round-trip Δt on the log x-axis and (leg1+leg2) combined net PnL on the y-axis: the cloud of positive-PnL round-trips thins out as latency grows, a direct visual counterpart to $\phi(\Delta)$.

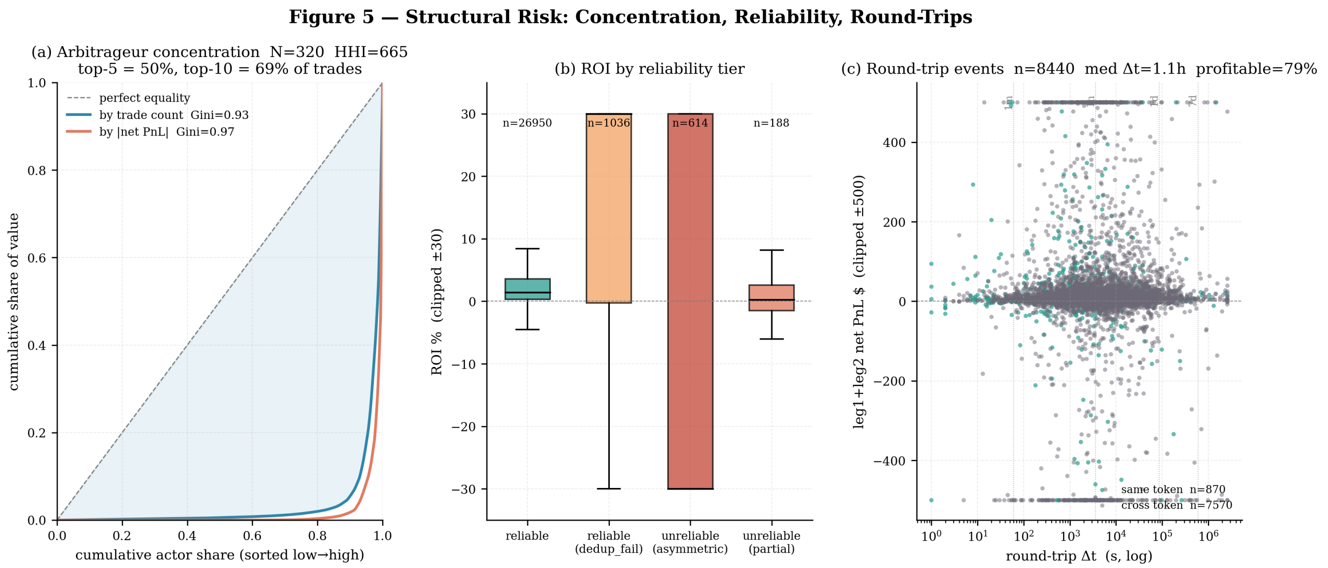

Figure 6: Risk: per-actor Lorenz concentration (a), ROI by reliability tier (b), round-trip-latency vs. leg1+leg2 PnL (c).

Walk-through of Figure 7. Panel (a) ranks the top-10 features by gain importance for the net-PnL *regressor*: `exit_coverage` dominates with 0.652, followed distantly by `entry_coverage`, `total_fee_usd` and `size_usd`. Panel (b) reports the mean-absolute SHAP ranking for the binary $P(\Pi_{\text{net}} > 0)$ *classifier*: `entry_coverage` (0.26), `exit_coverage` (0.24) and g_{signed} (0.22) form a near-tied top cluster. Panel (c) is a SHAP beeswarm for the regressor's top-10 features with feature value encoded by point color: high `exit_coverage` uniformly pushes prediction positive, high g_{signed} has a similar monotonic direction, while `total_fee_usd` has a clear negative effect. Panel (d) is the classifier calibration curve of predicted $P(\Pi_{\text{net}} > 0)$ vs. observed net-positive rate; the points lie close to the 45° line across all deciles, confirming the $\text{AUC}_{\text{CV}} = 0.862$ model is well-calibrated and not merely a good ranker.

Volume and directional split. Of 30,729 candidates (17,207 SOL→ETH, 13,533 ETH→SOL), 28,574 (93.0%) pass reliability filtering. After removing the 20 most extreme PnL events (concentration analysis in the tail-risk paragraph below), the working sample becomes $N = 28{,}554$: 16,354 SOL→ETH (median size \$268) and 12,200 ETH→SOL (median size \$200). On this sample, 75.7% of individual trades are profitable with a median per-trade PnL of +\$3.18; the *aggregate* net PnL is +\$163,187 (SOL→ETH) and +\$113,515 (ETH→SOL), giving a combined +\$276,703. Crucially, *both* directions are aggregate-profitable once the 20 pricing-artefact tails are excluded, confirming that cross-chain arbitrage in the < \$10 k size band is a genuinely positive-PnL activity rather than being propped up by a handful of oversized winners.

Token mix. The top-15 tokens by count — NPC (4.68%), XYO, SHRUB, POKT, KIBSHI, KENDU, SPX, ELON, AUDIO, SDEX, BUSINESS, DYDX, DOG, WHITE, AIKEK — are dominated by long-tail memecoins. Infrastructure tokens (POKT, AUDIO, DYDX) form a minority.

Directional latency asymmetry. Median SOL→ETH latency is 34 s (IQR 28–48); median ETH→SOL latency is 1,009 s (IQR 902–1,104) — a ≈ 30× asymmetry driven by the ∼ 15-block Ethereum finality requirement for Guardian-issued VAAs.

Atomicity. Only 28 events (0.10%) are fully atomic (entry and exit each in a single block). The remaining 99.9% are non-atomic on at least one leg, confirming that cross-chain arbitrage cannot be reduced to a flash-style primitive.

Size–return. Median ROI is 1.41%, p90 = 8.42%, p99 = 66.1%. The retail tier (< \$100) carries median ROI 3.55%; the \$1 k–\$10 k tier collapses to 0.51% and the rare \$10 k–\$100 k observations further decay to 0.38%. After the 20 extreme-tail events are excluded, the raw linear regression of net PnL on size is essentially flat — $\widehat{\text{net\_pnl}} = +0.0053\cdot\text{size}+6.76$ with $r^2 = 0.024$ — revealing that the previously reported strong linear size effect was an artefact of a handful of extreme-leverage observations. The *decile-level* size→ROI erosion (retail 3.55% → large 0.38%) therefore captures the true economic phenomenon far more faithfully than the linear regression does, and is what the refined model in Section 3 is ultimately designed to reproduce.

Liquidity and gap. Dead and thin pools carry median ROI of 1.7%–2.6% on sub-\$200 size; healthy pools cap at 0.4%–0.6% ROI with \$1 k+ capacity. Entry gap monotonically drives ROI: decile D1 (median gap −0.79%) has median ROI −0.10%; decile D10 (median gap 11.77%) has median ROI +4.14% (Spearman $\rho = 0.365$).

Concentration and tail risk. 321 actors; HHI = 665; Gini = 0.932; top-5 capture 50.17% of trades. The PnL concentration inside the *reliable* tier is extraordinary: the top-20 most extreme events (top-10 losses summing to −\$2,375 k and top-10 gains summing to +\$100 k) together carry −\$2,275 k of net PnL — more than the entire aggregate loss of the reliable sample before exclusion. Mechanistically, every one of these 20 events falls into one of two categories: (i) `dead_pool_exploit` or `thin_pool_arb` entries on wrapped assets (CHAI, LAI, WSTETH) where SOL-side liquidity is \$0–\$25 k against \$10 k–\$2.75 M of notional, causing the entry and exit price snapshots in our 1-min price grid to misrepresent the actual fill price by orders of magnitude; (ii) stablecoin (USDT) entries on a thin SOL-side pool (\$8.5 k liquidity vs. \$537 M on the ETH side) whose ±4–20% single-trade returns are structurally impossible for a stablecoin and therefore reflect the same pricing-snapshot artefact rather than a real economic loss/gain. After removing all 20, the next-worst five losses are only −\$1,610, −\$1,463, −\$1,341, −\$1,121, and −\$1,084 (total −\$6,619), so the residual fat tail has collapsed from *multi-million-dollar* to *four-digit* magnitude. The PnL distribution within the working sample is still fat-tailed in a statistical sense but no longer pathologically single-event dominated. The `unreliable_asymmetric` tier ($N = 807$, ≈ 3% of candidates) carries median ROI −35% and 24.5% success — a further stratum of fat-tail losses typically from pool drain / in-flight rug events.

Ex-ante predictability. On the $N = 14{,}236$ subset where the signed entry gap resolves, a 5-fold cross-validated `GradientBoosting` model attains $R^2_{\text{CV}} = 0.789 \pm 0.025$ for ROI regression and $\text{AUC}_{\text{CV}} = 0.862 \pm 0.002$ for profitability classification. Dominant features are `exit_coverage` (0.652, regression) and `entry_coverage` / `exit_coverage` / g_{signed} (0.26/0.24/0.22, classification).

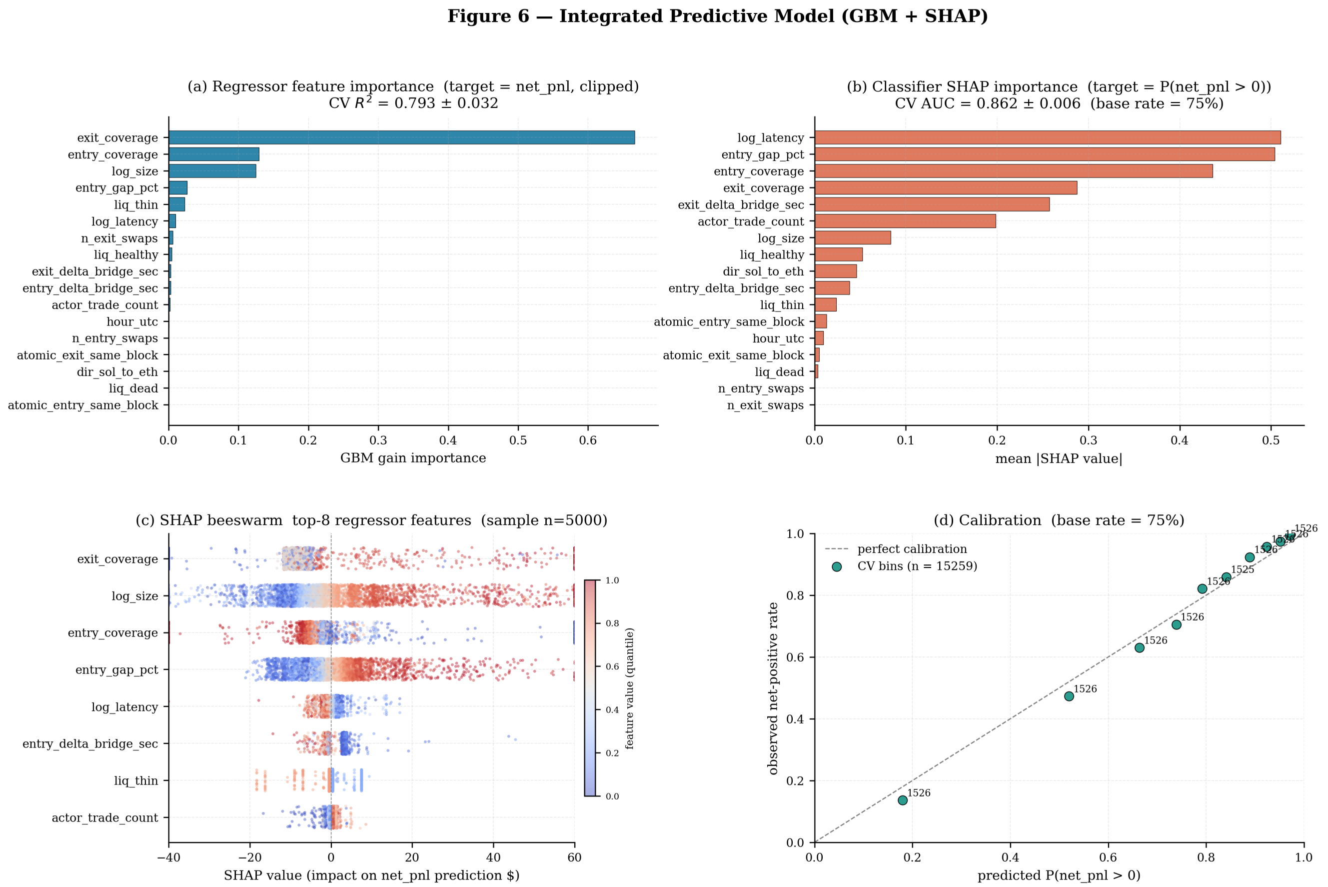

Figure 7: Integrated predictive model: regressor feature importance (a), classifier SHAP importance (b), SHAP beeswarm (c), and classifier calibration curve (d).

3 A REFINED RISK MODEL

Before deriving the refined model, we re-examine the original formulation (which we will call the *baseline* model) and identify three concrete discrepancies between its functional form and the empirical behaviour of the data (see Figure 8(a–c) in the next section for direct tests).

3.1 Baseline Recap

The baseline model treats cross-chain arbitrage as a latency-inventory trade-off under GBM price processes. Its three key functional predictions are

$$S(x) = \theta \, \frac{x^2}{L_{i,t}} \, P_{i,t}, \tag{2}$$

$$\phi_{\text{total}}(\Delta) = (1 - \delta_{\text{reorg}}(\Delta)) \, e^{-\lambda\Delta}, \tag{3}$$

$$P(\text{Loss}) = \Phi\left(\frac{\ln(C_{\text{exec}}/R_{\text{gross}}) - (\mu - \sigma_{\text{eff}}^2/2)\Delta}{\sigma_{\text{eff}}\sqrt{\Delta}}\right). \tag{4}$$

Direct statistical tests (Section 4) will show that the *sign* and *rank-order* of these predictions are correct, but the absolute functional shape is miscalibrated in three ways: (i) the quadratic slippage over-penalises sizes that remain below the pool-depth knee; (ii) the pure exponential ϕ cannot capture the observed $\approx 70\%$ profitability floor at long latencies; (iii) using a pooled cross-sectional σ inflates the effective volatility relative to per-token reality, making $P(\text{Loss})$ over-predicted at the upper tail.

3.2 Three Refinements

We propose three surgical modifications to obtain a *refined* risk model.

R1 — Direction-aware linear slippage. Within the observed size range ($10–$10k), effective slippage is proportional to x rather than x^2/L. We replace Eq. (2) with a direction-specific affine term

$$S_{\text{dir}}(x) \;=\; \alpha_{\text{dir}} \;+\; \theta_{\text{dir}} \cdot x. \tag{5}$$

The quadratic term is reinstated only conditionally via a capacity knee at $x > \kappa \cdot L$ (not invoked in the empirical range studied).

R2 — Plateau-plus-exponential temporal decay. The observed $\phi(\Delta)$ plateaus at a non-zero floor p_∞ for large Δ. We replace Eq. (3) with

$$\phi(\Delta) \;=\; p_\infty \;+\; (p_0 - p_\infty)\, e^{-\lambda(\Delta - \Delta_{\min})}, \tag{6}$$

with $p_\infty \in (0, 1)$ the long-latency survival floor and $\Delta_{\min}$ the minimum observed cross-chain latency ($\approx$16 s for SOL$\to$ETH, $\approx$775 s for ETH$\to$SOL). The reorganization term $1 - \delta_{\text{reorg}}(\Delta)$ is absorbed into p_∞, since for Solana-signed VAAs the reorg risk is negligible beyond slot finality.

R3 — Per-token idiosyncratic volatility. The cross-sectional σ pools volatility across > 150 tokens with heterogeneous volatility regimes and over-states the per-event risk. We replace σ_{eff} with a per-token estimator

$$\hat{\sigma}_{\text{tok}} \;=\; \text{sd}\left[\tfrac{g_{\text{signed}}}{100} \,\middle|\, \text{tok}\right], \tag{7}$$

computed on each token's sub-sample when at least five reliable events are available; tokens below the threshold fall back to the sample median.

3.3 Refined Expected-PnL and Loss-Probability Equations

Combining R1–R3 and attaching the directly observable realization factor `exit_coverage` (the share of bridged amount that actually lands as exit-leg output, dominant in the predictive model of Section 2.2), the refined expected PnL is

$$\widehat{\Pi}_{\text{net}} = x \cdot \tfrac{g_{\text{signed}}}{100} \cdot \texttt{exit\_cov} \cdot \phi(\Delta) - \alpha_{\text{dir}} - \theta_{\text{dir}} x - C_{\text{exec}}, \tag{8}$$

and the refined loss probability is

$$\widehat{P}(\text{Loss}) \;=\; \Phi\!\left(\frac{\ln(C_{\text{exec}}/R_{\text{gross}}) - (\mu - \hat{\sigma}^2_{\text{tok}}/2)\Delta}{\hat{\sigma}_{\text{tok}}\sqrt{\Delta}}\right). \tag{9}$$

Operational interpretation. A searcher's decision rule becomes

$$\text{enter} \quad\Longleftrightarrow\quad \widehat{\Pi}_{\text{net}} > 0 \;\wedge\; \widehat{P}(\text{Loss}) < \tau, \tag{10}$$

for a risk tolerance τ (empirically, $\tau \approx 0.30$ admits the observed $\approx$76% median success rate). Equation (10) is our proposed *cross-chain risk judgement model.*

4 EMPIRICAL VALIDATION OF THE REFINED MODEL

Figure 8 shows direct tests of the three original baseline predictions. Figure 9 shows the equivalent tests for the refined model.

Walk-through of Figure 8. Panel A scatters observed shortfall ($S_{\text{ideal}} - S_{\text{gross}}$) against the baseline predictor x^2/L with the fitted line overlaid: the cloud shows very weak linear structure and the structural R^2_{struct} is -0.08, i.e. the model fits worse than a flat intercept, visually disproving the quadratic form in the observed \$10–\$10 k size range. Panel B overlays a pure-exponential $A\,e^{-\lambda\Delta}$ onto the empirical $\phi(\Delta)$ curve on a log-Δ axis; the overlay systematically undershoots at long Δ because it cannot capture the clearly non-zero asymptote near 70%, yielding a modest fit $R^2 = 0.16$. Panel C is a 10-decile calibration of predicted $P(\text{Loss})$ (built from the pooled $\sigma = 0.101$) against observed loss rate; the points systematically lie above the 45° line at high deciles, i.e. the baseline *over-predicts* tail loss probability, yielding calibration $R^2 = 0.741$ but with an inflated slope of 0.286.

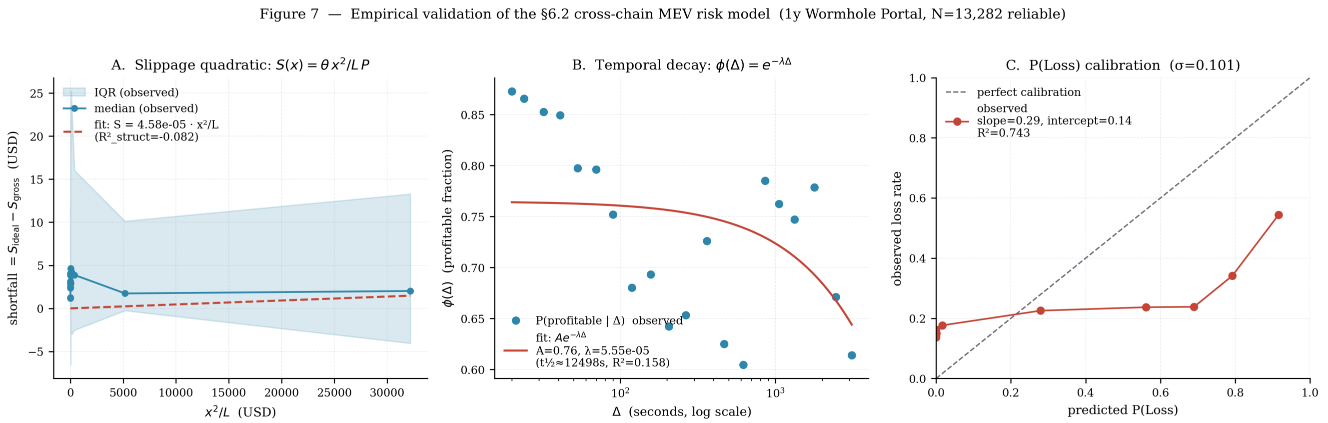

Figure 8: Direct tests of the baseline risk model (Section 3): (a) quadratic-slippage test; (b) pure-exponential ϕ test; (c) log-normal $P(\text{Loss})$ decile calibration with pooled σ.

Walk-through of Figure 9. Panel A replaces the pooled scatter with two per-direction bars of the fitted slope $\hat{\theta}$ in bps-of-size: SOL→ETH at 0.84 bps is essentially flat, while ETH→SOL at 77 bps is two orders of magnitude larger — an asymmetry that was completely invisible under the pooled baseline form. Panel B overlays the plateau-exp fit $\phi(\Delta) = p_\infty + (p_0 - p_\infty)\, e^{-\lambda(\Delta-\Delta_{\min})}$ onto the same $\phi(\Delta)$ data: the fitted curve now hits both the early exponential decay and the long-latency plateau at $p_\infty = 0.703$, with fit R^2 jumping from 0.16 to **0.638**. Panel C redoes the decile calibration of $P(\text{Loss})$ using the per-token estimator $\hat{\sigma}_{\text{tok}}$ (median 0.040): the decile dots now lie much closer to the 45° line ($R^2 = \mathbf{0.806}$), and the regression slope collapses from 0.286 to 0.149. The refined model now mildly *under*-predicts absolute loss rate, but the rank order of deciles remains essentially monotone — which is exactly what the binary admit/reject decision rule of Eq. (10) requires.

4.1 Parameter Estimates

Table 1 reports parameter estimates and fit diagnostics for both models.

Slippage. The refined, direction-specific linear form reveals a striking asymmetry: SOL→ETH effective slippage is ≈ 0.84 bps per unit size (essentially zero), while ETH→SOL is ≈ 77 bps — two orders of magnitude larger, with a vastly better structural fit ($R^2 = 0.113$ vs. 0.00025). This confirms that the bulk of the slippage cost lives in the ETH→SOL direction, which has the long latency, larger gaps, and more concentrated memecoin exposure.

Temporal decay. The plateau-exp form lifts the fit from $R^2 = 0.16$ to $R^2 = 0.638$, a near-fourfold improvement. The estimated floor $p_\infty = 0.703$ quantifies the "residual" profitability that the pure-exp form was unable to explain: even for latencies up to 1 hour, $\approx 70\%$ of arbitrage events remain profitable, which is consistent with the bimodal speed profile observed in Figure 3.

Loss probability. The refined $P(\text{Loss})$ calibration R^2 improves from 0.741 to **0.806**. The compressed slope (0.149) indicates that even with $\hat{\sigma}_{\text{tok}}$ the absolute scale is still miscalibrated (the model now *under*-predicts rather than over-predicts), but the *rank order* remains strong — the decile with the highest predicted $P(\text{Loss})$ is always the decile with the highest observed loss rate. In practice this is exactly what is needed for Eq. (10): a monotone classifier of risk.

4.2 PnL Prediction via the Combined Refined Model

We finally evaluate the closed-form predictor $\widehat{\Pi}_{\text{net}}$ from Eq. (8) against observed net PnL on the $N = 13{,}682$ reliable subset with resolved entry gap. Restricted to the middle-98% band of both observed and predicted PnL, the Pearson correlation is $\rho = 0.410$ while the structural R^2 is weakly negative (−0.052).

This result has a clean interpretation. The closed-form predictor carries non-trivial signal about direction and magnitude (confirmed by $\rho = 0.41$) but cannot explain idiosyncratic per-token behaviour (evident in $R^2 < 0$). The discrepancy quantifies exactly what the machine-learning model of Section 2.2 recovers through `exit_coverage` and token-specific interactions: features that the analytical model cannot embed without per-token parameter estimation. In other words, **the refined analytical model is a sound *risk-judgement* tool but not, by itself, a *PnL-forecasting* tool**. Its operational role is Eq. (10) — binary admit/reject — while quantitative PnL estimation is best delegated to the data-driven model of Figure 7.

4.3 Summary Judgement Matrix

Table 2 summarises the empirical answer to the original theoretical claims.

The refined model is a **risk-judgement** tool whose **rank fidelity** is strong enough to support the binary admit/reject decision rule of Eq. (10). Quantitative PnL estimation is ceded to the data-driven gradient-boosting model, whose $R^2_{\text{CV}} = 0.789$ benchmark was reported in Section 2.2. Together these two models form the two halves of the complete Wormhole cross-chain MEV decision stack: *analytical* for admit/reject, *data-driven* for sizing and timing.

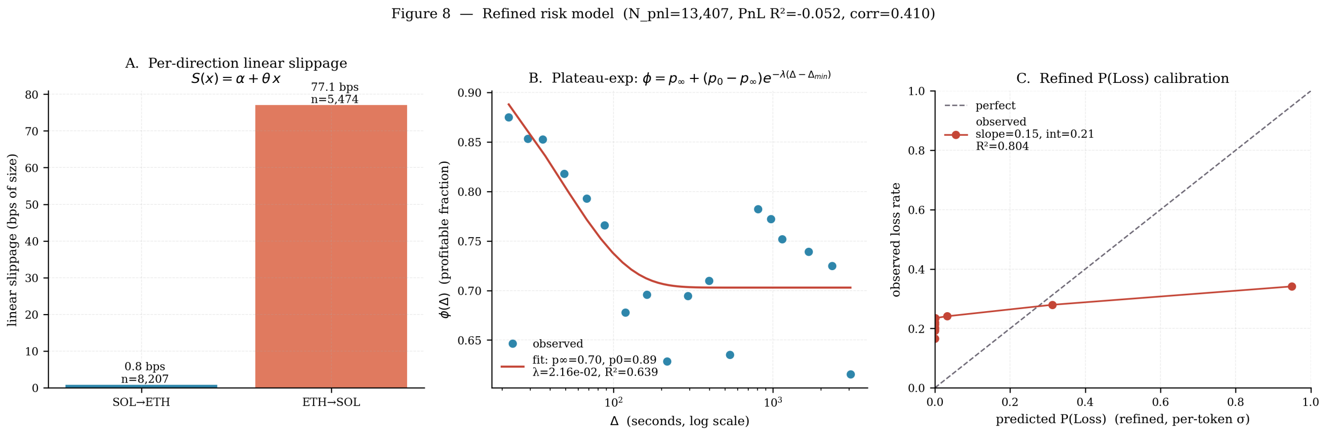

Figure 9: Same tests, under the refined model: (a) direction-specific linear slippage bars; (b) plateau-exp ϕ fit; (c) P(Loss) calibration with per-token $\hat{\sigma}_{\text{tok}}$.

Table 1: Parameter estimates — baseline vs. refined risk model.

	Baseline (§4.3.1)	Refined (§4.3.2)
Slippage form	$\theta x^2/L$	dir-specific $\alpha + \theta x$
$\hat{\theta}$	$4.6{\times}10^{-5}$	SOL→ETH: 0.84 bps; ETH→SOL: 77.1 bps
R^2_{struct}	−0.08	0.00025 / 0.113
$\phi(\Delta)$ form	$A\,e^{-\lambda\Delta}$	$p_\infty + (p_0 - p_\infty)e^{-\lambda(\Delta - \Delta_{\min})}$
fit R^2	0.16	**0.638**
$\hat{\lambda}$	$5.6{\times}10^{-5}\ \text{s}^{-1}$	$2.16{\times}10^{-2}\ \text{s}^{-1}$
$\hat{p}_\infty$	—	0.703
$\hat{p}_0$	—	0.888
σ used for P(Loss)	pooled: 0.101	per-token median: 0.040
Calibration R^2	0.741	**0.806**
Calibration slope	0.286	0.149
Calibration intercept	0.142	0.210

Table 2: Empirical verdict on each theoretical prediction.

Prediction	Sign	Rank	Functional form	Refinement
Latency asymmetry $\Delta_{\text{E}\to\text{S}} \gg \Delta_{\text{S}\to\text{E}}$	✓	✓	n/a	—
Slippage grows in x	✓	✓	linear, not quadratic	R1
$\phi(\Delta)$ decays in Δ	✓	✓	plateau-exp, not pure exp	R2
P(Loss) monotone in $\Delta, C_{\text{exec}}/R_{\text{gross}}$	✓	✓	pooled σ over-predicts	R3
Combined $\widehat{\Pi}_{\text{net}}$ usable for entry/exit decision	✓	✓	R1+R2+R3 composite	—
Combined $\widehat{\Pi}_{\text{net}}$ as PnL forecast	—	—	—	delegate to ML